

The well:

five faces of the ground state of the windowed Weil form, and the approximation lemma that would hold them

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6 September 2026, v2 (addendum §133: the lemma) — consolidates notebook §16, §66–68, §86–89, §120–127 and Grok’s lemma-theta-v-consolidated.md of riemann-weil-phenomenology

Abstract

On the window $[0, L]$, $L = \log \mu$, the truncated Weil form Q_L (archimedean part minus the prime towers up to μ) is positive with a smallest eigenvalue λ_0 that falls super-exponentially with μ — the *well*. Its ground state v has been measured from two sides over a week, by two independent chains (one on the space side, one on the zero side), on ζ , fifteen Dirichlet characters, eight elliptic curves and two Dedekind products, and the measurements fit one object: v is the unit-norm function of the window whose Fourier transform is hyper-null at the in-band zeros, whose spectral mass sits in the desert below the first zero, whose autocorrelation collapses super-exponentially at every lag beyond $y \approx 1.2$ (this is the “silence at the primes” of the quorum mechanism, which is not about primes), and whose value at the window edge is what remains: $\lambda_0 \approx \psi(0)^2 \cdot S$, with S the leakage of an edge jump onto the zeros beyond the band edge, computable from the zeros alone — the edge-doubling law $-\ln \lambda_0 = 2(-\ln |\psi(0)|) + O(1)$ found by Grok and verified on thirteen windows (the edge carries 82–103% of the depth). What is not established is the one thing that would turn the description into a depth law: how the minimal edge value depends on the configuration of in-band zeros. The desert alone does not fix it (dispersion by a factor 3 against $\tau\gamma_1$ on thirteen windows; the two-term formula and the one-set $\lambda_{\max}$ route are dead); the gap correction is $O(1)$ per cluster of zeros and its form is open. The note states the object, the evidence, the dead ends, and the approximation lemma. Nothing here bears on RH.

1 The object

Q_L on the hat/cosine basis $\{\eta_n\}_{n \leq N}$ of $[0, L]$; $\lambda_0 = \min Q_L$, v its eigenvector, $\psi = \sum v_n \eta_n$ the function, $\hat{v}$ its Fourier transform, $\Theta_v(y) = \int \psi(t)\psi(t+y) dt$ its autocorrelation, $\ell = -\ln \lambda_0$ the depth, $s = \ell/\mu$ the drilling speed, $\omega_{\max} = 2\pi N/L$ the band edge. Under RH, $Q_L(f) = \sum_{\gamma} |\hat{f}(\gamma)|^2$ over the zeros; the repository’s identity $Q_L = \text{zero Gram}$ holds to the tail on every L -function tried (zeros-from-the-radical, gl2-prime-side).

2 Five faces

1. The desert (§66–68, sampling-floor). λ_0 is the sampling constant of the zero set on the window, made small by the desert $[0, \gamma_1)$ where there are no zeros: under RH $c_L > 0$ (Theorem 1 of sampling-floor) and $\lambda_0(N) \downarrow c_L$. The Slepian cost of the desert alone, $1.71 L(\gamma_1 - \nu)_+/\mu$ per unit μ , is 60–90% of s on the fast characters and on the rank-0 elliptic curves.

2. Hyper-nullity and the edge budget (§16.3–16.7). $|\hat{v}(\gamma_k)|$ at the in-band zeros is exponentially smaller than any bound requires (4.7×10^{-72} at γ_1 for ζ , $\mu = 16$) — the extremity law $|\hat{v}(\gamma_1)| \approx C\lambda_0$; the budget $\lambda_0 = \sum_k |\hat{v}(\gamma_k)|^2$ is carried by the zeros at and beyond the band edge (leakage rate $\tau \approx 0.48$, §16.4).

3. The mass in the desert (§122). $|\hat{v}(\gamma)|^2$ is 0.8 at $\gamma = 1$, 0.25 at 5, 5×10^{-3} at 10, then $e^{-1.34\gamma}$ with hyper-null dips at the zeros, 10^{-60} at $\omega_{\max}$ (ζ , $\mu = 16$): the norm lives below γ_1 . Budget at the edge, mass in the desert — both true, distinct.

4. The collapse of the autocorrelation (§120–123; Grok’s space-side lemma). $\Theta_v(y) > 0$ everywhere, smooth (no band-edge oscillation at step 0.004), and $-\ln \Theta_v(y)$ grows super-exponentially: $\approx 0.2 s y e^y$ on ζ at $\mu = 16$ ($R^2 = 0.9995$, slope ratio $2.9 = (1 + y)e^y$, Gaussian excluded), with a coefficient $\propto 1/L$ across μ ($a \cdot L = 0.62, 0.62, 0.58$ at $\mu = 8, 11, 16$) and a shape that the window edge bends: steeper near L , gentler for composite objects. The space side (Grok) reads the same: ψ has a Gaussian bulk, $\psi(t) \approx \psi_{\text{mid}} e^{-a(t-L/2)^2}$ with $aL^2 = \ell$ on ζ , and steep edges (e^{-s}); the $y e^y$ tail of Θ_v is the edge, not the desert weight. The towers sample this collapse at the prime lags: $-\ln \delta_p \approx 0.19 s p \log p$ (one $\Gamma_{\mathbb{R}}$) or $0.38 s p \log p$ (several; degrees 2 and 3 alike, §118–119) is the same law read at $n = p^k$, and at lags with no prime the same numbers appear (§120–121). Consequence for the quorum mechanism: its first half (the ground state is deaf to each prime) is a generic property of the well; only the coupling κ_p , Gaussian in $(p \log p)^2 / (\mu \log \mu)$, is arithmetic. The ground state’s sum rule sees only 2 and 3 (98.9% + 1.1% for ζ).

5. The edge value (Grok; §125, §127). $-\ln \lambda_0 = 2(-\ln |\psi(0)|) + R$, $\psi(0) = L^{-1/2}(v_0 + \sqrt{2} \sum_{n \geq 1} v_n)$: ratio 2.06–2.19 on ζ , χ_3 , χ_{29} , $\zeta L(\chi_3)$; $-2 \ln |\psi(0)|/\ell \in [0.82, 0.98]$ on Grok’s eleven windows and 0.97–1.03 on ζ — thirteen windows, five orders of magnitude in λ_0 , the edge carries the depth. And why: a function with value $\psi(0)$ at both ends has a jump whose transform is $2\psi(0) \sin(\gamma L/2)/\gamma$; in band the smooth part cancels it (hyper-nullity), beyond the band it does not. On ζ at $\mu = 11$ with 500 zeros, the zeros beyond $\omega_{\max}$ carry 93.3% of λ_0 and the estimated tail beyond 811 completes it to 1.5%; the jump prediction $\sum_{\gamma_k > \omega_{\max}} 8\psi(0)^2 \sin^2(\gamma_k L/2)/\gamma_k^2$ matches to a factor 1.55 in the sum, predicting $R = 3.37$ against 3.29 measured. Hence

$$\lambda_0 \approx \psi(0)^2 \cdot S, \quad S = \sum_{\gamma_k > \omega_{\max}} \frac{8 \sin^2(\gamma_k L/2)}{\gamma_k^2} \times O(1.5) \approx \frac{4}{\pi} \frac{\log(\omega_{\max}/2\pi e)}{\omega_{\max}},$$

which explains why R depends on N (through $\omega_{\max}$) and on the object (through the zero density), and why it changes sign when in-band zeros near the band edge still carry budget (ζ , $\mu = 16$: $R = -4.4$).

3 One object

The five faces are one statement: *the ground state is the unit-norm function of the window that is hyper-null at the in-band zeros and minimizes its value at the edge; its mass sits in the desert because that is where no constraint holds; its autocorrelation collapses because a smooth bump with steep edges has one; and the depth is the square of the minimal edge value times the leakage of the edge jump onto the zeros beyond the band.* The primes enter twice, and only twice: as the samples of the collapse (harmless), and in the coupling κ_p that interlocks the towers (the arithmetic of the quorum).

4 What is dead

The two-term formula $-\ln c_L \approx aL(\gamma_1 - \nu)_+ + bL \sum (\text{gap} - \nu)_+$ (its gap sum does not converge; at a common cutoff it overpredicts most L -functions, §92–94). The one-set constant by $\lambda_{\max}(\chi_E P \chi_E)$ ($\lambda_{\max}$ sees only the largest interval; Grok, lemma2-sampling-constant). The line $\ell \approx a\tau\gamma_1 + b$ (jackknife: the slope halves without the one deep window). The Gaussian form of the collapse in y (slopes). The reading of the silence as arithmetic (§120). Ten pre-registered executions died on this object in one week; the object survived all of them.

5 What is open: the approximation lemma

Lemma 1 (to prove). *Let $\Gamma_L = \{\gamma_k < \omega_{\max}\}$ be the in-band zeros. Among unit-norm functions on $[0, L]$ whose Fourier transform vanishes on Γ_L (to the hyper-null order observed), the minimal edge value satisfies $-\ln |\psi(0)|_{\min} = \frac{1}{2} s \mu + O(1)$, i.e. $\lambda_0 \asymp \psi(0)_{\min}^2 S(\omega_{\max})$, with s a functional of the configuration Γ_L : $s = C_0 \tau \gamma_1 / \mu + (\text{gap correction})$, the correction $O(1)$ per cluster of zeros.*

What the data say about s : the desert alone gives a factor-3 dispersion ($\ell / \tau \gamma_1 \in [2.3, 7.4]$ on thirteen windows, ζ the deepest per unit of desert); the gap correction is $O(1)$ per piece in the log-determinant of the one-set compression (Widom's perimeter term: 0.46–0.62 nats per gap, position-independent), which matches ℓ to 20% on χ_5 only; no elementary (γ_1, L) predictor of $|\psi(0)|$ exists. The lemma is pure harmonic analysis — an extremal problem with linear constraints on a window — and it contains no prime. Its proof would give the depth law; its failure would tell which constraint the picture misses.

Addendum (notebook §133): the lemma, found

The eigenvalues of the zero Gram on the window form a *ladder*: $-\ln \lambda_k = 107.5, 91.4, 76.9, \dots, 4.9$ then $O(1)$ for ζ at $\mu = 11$ — exactly $d = 10$ exponentially small eigenvalues, with rungs 16.1, 14.5, 13.4, $\dots$, 4.2 whose sum is ℓ . The rungs are the same across objects (16.1, 16.1, 14.5 first rungs on ζ, χ_3, χ_5); what differs is their number: 10, 5, 3, 1 for $\zeta, \chi_3, \chi_5, \chi_{29}$, and that number is $d = \lceil \max_{\gamma} (\gamma L / 2\pi - N_{\Gamma}(\gamma)) \rceil$ — the maximal lead of the window's Nyquist count over the zero count (a discrete Landau theorem: zeros above the Nyquist crossing free nothing). Hence $\ell \approx \bar{r} d$ with $\bar{r} \approx 11$ nats, the mean rung: on twelve windows with ℓ from 8.5 to 140, $\ell/d = 10.9, 11.9, 11.1, 11.6, 9.9, 11.9, 11.3, 11.0, 9.9, 10.7$ (outliers: $d = 1$, and sub-Nyquist ζ at $\mu = 16$ where d grows with N while ℓ saturates). No fitted parameter but $\bar{r}$; d is read off the zero list and N . The two-term formula is replaced by a count: *the depth is eleven nats per dimension the zeros leave free below the window's Nyquist line*. The lemma to prove is now (A) the discrete Landau count and (B) the universality of the plunge ladder $16 \rightarrow 5$.

Status

Every number above is measured, with its script and its judge in the repository; the identity $Q_L = \text{Gram}$ is the only theorem used and it is used under RH. The two chains — space side (Grok: $\psi, aL^2 = \ell$, edge doubling) and zero side (this note: budget, mass, collapse, leakage) — were run independently and agree. Nothing here bears on RH.